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Research article

# A machine learning model for prediction of 30-day primary graft failure after heart transplantation

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ABSTRACT

*Background:* Primary graft failure (PGF) remains the most common cause of short-term mortality after heart transplantation. The main objective was to develop and validate a risk model for prediction of short-term mortality due to PGF after heart transplantation using the ISHLT Heart Transplant Registry.

*Methods:* We developed a non-linear artificial neural networks (ANN) model to evaluate the association between recipient-donor variables and post-transplant PGF. Patients in the ISHLT registry were randomly divided into derivation and an independent internal validation cohort. The primary endpoint was PGF defined as death within 30 days due to Graft failure or Cardiovascular causes or retransplant within 30 days for causes other than rejection.

*Results:* Among 64,964 adult recipients transplanted between 1994 and 2013, mean age was 51 years and 22% were female. The incidence of PGF up to 30 days was 3.7%. The ANN model selected 33 of 77 risk variables as relevant for PGF prediction. The C-index in the test cohort was 0.70 (95% CI: 0.68-0.71). The risk variables which most influenced the PGF were underlying HF diagnosis, ischemia time and sex, while renal function had a lower influence.

*Conclusion:* An ANN model to predict primary graft dysfunction was derived and independently validated. The good discrimination of the ANN model likely results from its flexibility to model potentially non-linear relationships and interactions. Whether this model with improved discrimination can assist in clinical decisions at the time of transplant should be tested.

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## 1. Introduction

Primary graft failure (PGF) is the leading cause of 30-day mortality after heart transplantation, and also contributes to prolonged mechanical ventilation and ICU stay [1]. Although the etiology of PGF is poorly understood, preexisting donor heart disease, ischemia during the organ preservation process, reperfusion injury immediately after re-implantation, and recipient risk factors and their interaction with the donor characteristics contribute to the development of PGF. Due to a shortage of donor hearts, some transplant programs decided to accept hearts from "suboptimal" donors, eg, patients over age 55, those with LV hypertrophy or non-obstructive coronary artery disease [2]. Therefore, an accurate prediction of PGF occurrence is becoming even more important.

Currently, classical statistical methods are the most common way to identify and rank variables of clinical importance. Although many risk factors associated with post-transplantations survival have been identified, the relationships among these variables (interactions) have not been fully explored [3,4]. Interactions among risk factors are complex and may be donor/recipient specific as suggested in a previous study of post-transplant survival [5]. In addition, pre-operative risk factors might influence the outcome in a non-linear way, making conventional statistical methods such as multivariable linear, logistic or proportional hazards regressions inadequate for assessing post-transplant outcomes. For example, the RADIAL score which was designed to predict the risk of PGF by computing a score of six independent risk factors, was obtained with multivariable logistic regression analysis [6].

Neural network modeling is an emerging approach that focuses on learning relationships from data. It allows a more flexible relationship between variables and outcomes and can search for the best fit by identifying patterns in the data. Studies have demonstrated its feasibility in risk prediction [7,8], especially when a large set of risk factors are believed to interact with one another and potentially affect the outcome interactive fashion [9].

Thus, the aim of this study is to establish a neural network model for risk prediction of PGF within 30 days after heart transplantation.

## 2. Material and methods

### 2.1. Study population

Data of heart donors and the corresponding recipients transplanted between January 1, 1994, and December 31, 2013, were collected from the ISHLT registry (n = 84,325 donor-recipient pairs). Pediatric cases (recipients younger than 18 years, n = 9270 or donor age <15 years, n = 1870) and those with incomplete mandatory data (diagnosis, blood group, age, gender, duration of follow-up, and/or cause of death, n = 7426) were excluded. The final study population comprised 65,759 transplantations and 64,964 patients.

### 2.2. Data source

The ISHLT registry is a global collaborative effort and the largest repository of heart transplant data in the world [10]. The registry includes data stored since 1983 and contains comprehensive recipient pre-transplantation, transplantation, discharge, and follow-up variables as well as donor data. Post-transplant information is reported at the end of the annual follow up period and at the time of death.

### 2.3. Study design

The features used in the model were based on variables available at the time of transplantation (N = 52). Features including more than two categories were dichotomized to category-1 variables, yielding a total of 83 variables used in the modeling. The primary endpoint was primary graft failure defined as death within 30 days due to Graft failure, due to Cardiovascular causes, or retransplant within 30 days for causes other than rejection. The Ethics Committee for Clinical Research at Lund University, Sweden approved the study protocol.

### 2.4. Missing data

The variables represent a combination of data with mandatory submission and data with optional submission. The data missingness was ranged from 0.25% to 71% (mean 42%). We evaluated two different imputation techniques, where the first one was multiple probability imputation (MPI) during training of the models and mean imputation during validation. The second approach was simple mean imputation (MI) for continuous variables and using an extra category for binary and multi-categorical variables. The latter approach can be biased if data are not missing at random. All imputations were stratified by time era.

### 2.5. Neural network model and deep learning

We consider only fully connected neural network architectures (MLP) with up to three hidden layers. For a general introduction to neural networks see Bishop et al., *Neural Networks for Pattern Recognition* [11]. The number of hidden nodes in each layer is considered as a hyper parameter, with allowed values between 25 and 100. Hidden units all use either rectified linear (ReLU) or standard hyperbolic tangent activation functions. The output uses sigmoid activation to obtain an event probability. Zero layers essentially

degrades the network to plain logistic regression, which was also tested as reference. Regularization methods considered were L2 weight decay, L2 max norm and dropout [12,13]. Each of the regularization methods were considered as a hyper parameter allowing for all methods to be present at the same time.

The MLP models were trained by standard stochastic gradient descent techniques using the standard cross entropy loss function. As the dataset was unbalanced, with only 3% empirical probability of an event, a weighted error function was used. Positive samples were given a weight of three compared to negative samples that had unity weights. During evaluation of the models, equal weights were used. The data was pre-processed before modeling such that continuous variables were normalized to zero mean and unit variance, binary values were represented as 0 and 1, and categorical variables were transformed to a "one-hot" representation. The model selection strategy was based on stratified 5-fold cross validation. The model selection found the best set of parameters using random search [14]. Table 1 lists all hyper parameters and their sampling distributions.

### 2.6. Statistical analysis

The discriminatory power of the neural network models for 30-day PGF was evaluated by calculating the area under the receiver operating curve (AUROC). To compare different areas and compute confidence intervals a bootstrap method was used as implemented by the pROC R-package [15]. The model validation strategy was internal using a nested K-fold cross-validation approach (Fig. 1). An outer 5-fold cross-validation loop divided the entire data into five sets of equal size, where each set was used as a test set in turn. The remaining data were used to train the MLP model. For each such training set model selection (i.e. finding suitable hyperparameters) was accomplished using an inner 5-fold cross-validation loop. All splits were stratified. All test predictions, from the 5 test sets, were in the end concatenated to one final test set comprising all cases [7]. Odds ratios were calculated overall cases, by computing it for every test set using its complimentarily trained model and taking the average (geometric mean). For continuous variables, the actual value was compared against the value incremented by one unit. For binary values, the positive value was compared against the negative value. For categorical variables, odds ratios were calculated using one of the categories as base. For univariate descriptive statistics, Mann-Whitney U-tests were performed for continuous variables and $\chi^2$ tests were used to compare categorical variables.

To further assess the importance of individual variables for prediction performance in the model, we performed a randomization test. For a specific variable, we randomly permuted all values in that column. If the variable is important, this should lead to a drop in AUROC score. For each variable, we repeated this randomization 100 times and calculated the mean and standard deviation. Furthermore, we calculated the Z-score for the true AUROC value (using the correct column values) under this empirical distribution. For unimportant variables, the Z-scores will follow a standard normal distribution, hence large Z-scores will be an indicator of importance. Furthermore, a regression tree was constructed on the full dataset using the odds calculated (geometrical mean) by the best MLP model as the outcome, and all covariates as regressors. Categorical variables were all transformed into one-hot binary variables, such that any decision will be a specific value against the rest. In particular, the gender information is coded as an indicator for each of the four possible gender combinations between the recipient and the donor. This analysis shows the relative importance of the variables as selected by the model for predicting the final risk, and in addition the typical threshold values for the continuous variables.

### 2.7. The RADIAL model

We compared the performance of the neural network model with the RADIAL method [6]. This RADIAL method computes a score based on the variables Right atrial pressure, recipient age, presence of diabetes mellitus, inotropic dependence, donor age and length of ischemic time. A high score is predictive of PGF. The variable right atrial pressure was not available in our cohort, instead we used the surrogate variable pulmonary artery pressure (diastolic) divided by two. The RADIAL method was evaluated using the score as defined in Segovia et al. and by recalibrating a logistic regression model as to adapt it to the present cohort and the replacement of one predictor variable.

### 2.8. Software

All models were implemented using Theano and Lasagne. Scikit-learn was used for some functionality such as AUROC calculations [16–18].

**Table 1**
Hyperparameters of the MLP models and their sampling distribution used during model selection.

| Parameter | description | Sampling distribution |
|---|---|---|
| Nh | Number of hidden nodes | Choice [25, 50, 100] |
| Λ | L2 regularization strength | LogUniform [$10^{-5}$, $3*10^{3}$] |
| P | Dropout probability | Choice [0, 0.1, 0.2] |
| Mn | Maxnorm (per unit) | LogUniform [0.02, 0.3] |
| Act | Activation function | Choice [relu, tanh] |

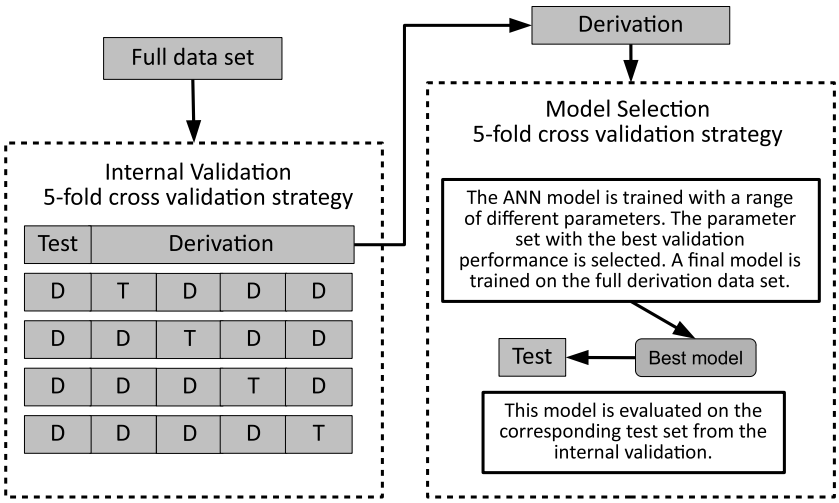


**Fig. 1.** Description of the validation and model selection procedure. The full data cohort is randomly divided into five parts of equal size, stratified on PGF status. Each part is used as a test set while the remaining is used for training. Each training session is also using stratified 5-fold cross validation to find optimal model parameters and the resulting model is then tested on the corresponding test set.

## 3. Results

### 3.1. Characteristics of the data cohort

The data cohort consisted of 65,759 patients that have undergone heart transplantation. The incidence of PGF was 3.7% (N = 2465). The mean age was 51.4 ± 11.9 years (median 54, range 18–79 years), and 22% (N = 13,607) of the transplanted patients were female. A complete listing of all variables is shown in Table 2 together with univariate descriptive statistics. Approximately 90% (N = 46) of the variables have missing data, ranging from 0.25% to 71% (mean 42%). The degree of missing data is larger for the PGF cases (mean 57%) compared to the control cases (mean 42%).

Table 2 lists donor and recipient characteristics among those with and without PGF. Significant differences were present among clinical characteristics between the two groups. For example, PGF was more likely in female-to-male donor-recipient sex matches compared to the other sex-match combinations. Donor-recipient weight match was not significantly associated with PGF. Donor hypertension, cocaine history, and donor death from stroke were also more likely in the PGF group.

### 3.2. Characteristics of the neural network models

When comparing different network architectures, no advantage could be found using more than one hidden layer. The most common size for the hidden layer was however not the smallest possible, indicating an advantage of a more complex architecture. When comparing the choice of activation functions, a clear advantage in performance was found using ReLU.

### 3.3. Validation of the model

Table 4 presents the average validation and test AUROC performances for one- and two-layer MLPs, together with logistic regression (0 hidden layers MLP). Here validation AUROC refers to the average AUROC values obtained during model selection and test AUROC refers to the average AUROC values obtained when testing models on each of the five independent test sets (see Fig. 1). We also tested different imputation schemes. For random imputation, the best performance was obtained using a one hidden layer network with an AUROC of 0.69 (95% CI 0.67, 0.69) and an area under precision-recal curve of 0.082. The linear model obtained a corresponding AUROC of 0.67 (CI 0.66, 0.68), p-value = 0.002 with an area under precision-recal curve of 0.076. No improvement could be found using networks with more than one hidden layer. For mean imputation, including a missing value marker on categorical variable, an improvement in performance was found. This was most noticeable for the linear network with an improvement of about 3% (66.7% vs 69.9%). This highlights the difficulty of using mean imputation when there is a correlation between PGF and the amount of missing data, resulting in biased test AUROC values. Overall, the differences between validation and test AUROC performance values were small, indicating that possible overtraining difficulties were kept to a minimum.

### 3.4. Variable ranking

Using the best performing model, all variables with a rank value (Z-score) larger than 2 are presented in Table 3, representing significantly important variables under the permutation test. In addition, efficient odds ratios (EOR) (geometric mean and 2.5%, 97.5%

**Table 2**
Baseline donor and recipient characteristics at the time of transplant.

| Variable | Control | | PGF within 30 days | | |
|---|---|---|---|---|---|
| | Data | Missing (%) | Data | Missing (%) | p-value |
| | N = 63,294 | | N = 2465 | | |
| **Recipient data** | | | | | |
| Female sex | 13,607 (21.5%) | 0 | 536 (21.7%) | 0 | 0.789 |
| Age | 51.4 ± 11.9 | 0 | 52.1 ± 11.6 | 0 | <0.001 |
| Weight | 78.9 ± 16.6 | 23.4 | 77.3 ± 16.5 | 36.5 | <0.001 |
| Height | 173.8 ± 9.5 | 23.9 | 172.3 ± 9.5 | 37.0 | <0.001 |
| Diagnosis | | 0.3 | | 0.1 | <0.001 |
| NICM | 30,790 (48.8%) | | 1025 (41.6%) | | |
| ICM | 24,740 (39.2%) | | 1035 (42.0%) | | |
| Misc | 3267 (5.2%) | | 125 (5.1%) | | |
| Valve | 1767 (2.8%) | | 120 (4.9%) | | |
| Congenital | 1304 (2.1%) | | 83 (3.4%) | | |
| Graftfailure | 1265 (2.0%) | | 74 (3.0%) | | |
| PVR | 2.5 ± 1.9 | 52.7 | 2.8 ± 2.1 | 69.3 | <0.001 |
| Chronic obstructive pulmonary disease | 1262 (3.8%) | 47.5 | 31 (3.4%) | 62.6 | 0.544 |
| Smoke history | 8584 (46.0%) | 70.5 | 179 (39.8%) | 81.7 | 0.01 |
| Hypertension | 14,922 (41.2%) | 42.7 | 423 (41.6%) | 58.7 | 0.829 |
| Vascular disease | 1149 (1.82%) | 44.0 | 50 (5.0%) | 59.4 | 0.003 |
| CVI at listing | 1597 (4.3%) | 41.1 | 42 (4.1%) | 58.7 | 0.866 |
| Stroke after listing | 904 (2.8%) | 48.5 | 35 (3.9%) | 64.0 | 0.048 |
| Insulin treated diabetes | 620 (1.7%) | 40.7 | 18 (1.8%) | 58.5 | 0.889 |
| Infection within 2 weeks | 3609 (10.0%) | 43.0 | 110 (11.5%) | 61.1 | 0.15 |
| Dialysis | 1247 (3.6%) | 44.7 | 68 (7.2%) | 61.8 | <0.001 |
| Defibrillator | 18,274 (45.3%) | 47.2 | 432 (47.6%) | 63.2 | <0.001 |
| Antiarrhythmics | 12,690 (37.3%) | 46.2 | 418 (43.7%) | 61.2 | <0.001 |
| Amiodarone | 8956 (27.4%) | 48.3 | 289 (31.6%) | 62.8 | 0.006 |
| Most recent creatinine | 1.4 ± 1.6 | 45.7 | 1.6 ± 1.5 | 61.5 | <0.001 |
| Bilirubin | 1.3 ± 2.7 | 45.1 | 1.8 ± 4.9 | 62.9 | <0.001 |
| Albumin | 36.8 ± 7.3 | 65.2 | 36.5 ± 7.5 | 79.2 | 0.942 |
| Medical condition | | 40.5 | | 57.7 | <0.001 |
| Home | 18,225 (48.4%) | | 446 (42.8%) | | |
| ICU | 13,373 (35.5%) | | 444 (42.6%) | | |
| Hospital | 6082 (16.1%) | | 152 (14.6%) | | |
| Inotropic support pre-transplant | 17,434 (46.4%) | 40.6 | 470 (45.9%) | 58.5 | 0.79 |
| Ventilator | 933 (2.5%) | 41.3 | 108 (10.9%) | 59.6 | <0.001 |
| ECMO | 163 (0.4%) | 41.4 | 37 (3.7%) | 59.7 | <0.001 |
| Intra-aortic ballon pump | 2135 (5.7%) | 41.1 | 98 (9.7%) | 59.2 | <0.001 |
| Prior transplantation | 1558 (2.5%) | 3.4 | 114 (4.8%) | 3.4 | <0.001 |
| Prior cardiac surgery | 13,965 (35.5%) | 37.8 | 442 (40.5%) | 55.7 | <0.001 |
| Cytomegalo virus status positive | 16,885 (72.1%) | 63.0 | 451 (77.2%) | 76.3 | 0,007 |
| Pre-transplant blood transfusion | 13,276 (48.6%) | 56.8 | 492 (63.2%) | 68.4 | <0.001 |
| PRA >10% | 4620 (14.0%) | 47.7 | 154 (17.8%) | 64.8 | 0.002 |
| Transplant era | | 0 | | 0 | <0.001 |
| 1996–2000 | 16,405 (25.9%) | | 792 (32.1%) | | |
| 2006–2010 | 15,593 (24.6%) | | 567 (23.0%) | | |
| 2001–2005 | 15,005 (23.7%) | | 506 (20.5%) | | |
| 2011–2015 | 9879 (15.6%) | | 363 (14.7%) | | |
| 1991–1995 | 6412 (10.1%) | | 237 (9.6%) | | |
| **Donor data** | | | | | |
| Age | 34.1 ± 12.7 | 0 | 38.3 ± 12.8 | 0 | <0.001 |
| Ischemic time (min) | 189.0 ± 64.9 | 28.2 | 203.4 ± 75.5 | 42.4 | <0.001 |
| Cytomegalo virus status positive | 23,156 (60.1%) | 39.1 | 618 (59.8%) | 58.1 | 0.888 |
| Hypertension | 4801 (12.6%) | 39.6 | 182 (17.5%) | 57.8 | <0.001 |
| Diabetes mellitus | 896 (2.3%) | 39.5 | 27 (2.6%) | 57.6 | 0.676 |
| Left ventricular ejection fraction | 61.6 ± 7.5 | 60.5 | 61.8 ± 8.1 | 77.0 | 0.734 |
| Inotropic support | 18,168 (46.5%) | 38.2 | 485 (46.9%) | 58.0 | 0.821 |
| Bilirubin | 1.1 ± 1.5 | 46.7 | 1.1 ± 1.7 | 65.4 | 0.024 |
| Creatinine most recent | 1.3 ± 1.5 | 48.4 | 1.3 ± 1.5 | 65.2 | 0.409 |
| BUN | 15.3 ± 12.6 | 44.2 | 15.9 ± 14.0 | 63.7 | 0.554 |
| Smoke history | 10,315 (28.2%) | 42.2 | 330 (34.1%) | 60.7 | <0.001 |
| Cocaine history | 3527 (9.7%) | 42.7 | 65 (6.8%) | 61.2 | 0.003 |
| CODD | 20.1% | | 28.4% | | <0.001 |
| Head trauma | 25,365 (50.1%) | | 669 (37.9%) | | |
| Stroke | 13,845 (27.4%) | | 711 (40.3%) | | |
| Other | 11,386 (22.5%) | | 385 (21.8%) | | |
| **Donor-recipient data** | | | | | |

(*continued on next page*)

**Table 2** (*continued*)

| Variable | Control | | PGF within 30 days | | |
|---|---|---|---|---|---|
| | Data | Missing (%) | Data | Missing (%) | p-value |
| | N = 63,294 | | N = 2465 | | |
| HLA-A | | 52.2 | | 63.4 | 0.049 |
| 2 mismatch | 16,824 (55.6%) | | 502 (55.7%) | | |
| 1 mismatch | 11,928 (39.4%) | | 339 (37.6%) | | |
| 0 mismatch | 1489 (4.9%) | | 60 (6.7%) | | |
| HLA-B | | 48.6 | | 63.4 | 0.132 |
| 2 mismatch | 23,603 (72.6%) | | 681 (75.6%) | | |
| 1 mismatch | 8348 (25.7%) | | 206 (22.9%) | | |
| 0 mismatch | 580 (1.8%) | | 14 (1.6%) | | |
| HLA-DR | | 53.2 | | 65.1 | 0.961 |
| 2 mismatch | 17,230 (58.2%) | | 504 (58.5%) | | |
| 1 mismatch | 11,070 (37.4%) | | 318 (36.9%) | | |
| 0 mismatch | 1314 (4.4%) | | 39 (4.5%) | | |
| Blood group | | 0 | | 0 | <0.001 |
| Match | 46,925 (74.1%) | | 1925 (78.1%) | | |
| Compatible | 16,319 (25.8%) | | 531 (21.5%) | | |
| Incompatible | 50 (0.1%) | | 9 (0.4%) | | |
| Gender | | 0 | | 0 | <0.001 |
| Match | 45,825 (72.4%) | | 1591 (64.5%) | | |
| Female to male | 11,629 (18.4%) | | 674 (27.3%) | | |
| Male to female | 5840 (9.2%) | | 200 (8.1%) | | |
| Height | | 25.9 | | 39.5 | 0.588 |
| 0.95–1.05 | 29,073 (62.0%) | | 896 (60.1%) | | |
| 0.90–0.95 | 7780 (16.6%) | | 250 (16.8%) | | |
| 1.05–1.10 | 5686 (12.1%) | | 194 (13.0%) | | |
| <0.90 | 2293 (4.9%) | | 78 (5.2%) | | |
| >1.10 | 2083 (4.4%) | | 73 (4.9%) | | |
| Weight | | 23.9 | | 37.4 | 0.324 |
| 0.85–1.13 | 25,854 (53.7%) | | 823 (53.3%) | | |
| 1.13–1.30 | 7798 (16.2%) | | 242 (15.7%) | | |
| 0.70–0.85 | 7381 (15.3%) | | 234 (15.2%) | | |
| >1.13 | 4325 (9.0%) | | 162 (10.5%) | | |
| <0.70 | 2780 (5.8%) | | 82 (5.3%) | | |

Data are mean ± SD or number (%). The numbers are calculated from patients with available data. Variables included in: BUN, blood urea nitrogen; CODD, cause of donor death; CVI; ECMO, extracorporeal membrane oxygenation; HLA, human leucocyte antigen; ICM, ischemic cardiomyopathy; ICU, Intensive care unit; NICM, non-ischemic cardiomyopathy; PRA, panel reactive antibody level; PVR, pulmonary vascular resistance. Tests used: Pearson test or Kruskal-Wallis test.

percentile range values) are also presented in Table 3. A variable with an efficient odds ratio range either below one or above one is showing a consistent behavior across the whole data cohort. The Z-score ranking found donor age to be the most important variable, with an efficient odds ratio of 1.22 (unit change = 11 years). The recipient diagnosis, as a group, was also found to be important, where the diagnoses congenital heart disease and Valvular Cardiomyopathy having efficient odds ratios of 1.7 and 1.6, respectively. The third place in the Z-score ranking list was the need for mechanical ventilation at transplant (EOR = 1.9). Sex match was also important, with female donor to male recipient associated with a higher risk of mortality (EOR = 1.34). The fifth most important variable was donor Ischemic time where an increase of 45 min resulted in an increased risk (EOR = 1.12) followed by the renal function with a EOR of 1.09.

### 3.5. Regression tree interpretation

Fig. 2 shows the regression tree that was fitted to the predictions, represented by the odds calculations of the best MLP model. The depth of the regression tree was constrained to four levels to only focus on the most important decisions. The age of the donor is considered particularly important, as well as gender match of female donor to male recipient.

### 3.6. Comparison with the RADIAL method

The score based RADIAL model obtained an AUROC of 0.53 (CI 0.52, 0.54), which is significantly lower that our best neural network mode ($p < 0.001$). The RADIAL model was also recalibrated on our data cohort which resulted in an AUROC of 0.61 (CI 0.60, 0.62), also lower than the neural network-based model ($p < 0.001$). The RADIAL model was also tested without surrogate variable for the right atrial pressure resulting in an AUROC of 0.53 (CI 0.52, 0.54). A corresponding recalibration gave an AUROC of 0.61 (CI 0.59, 0.62).

**Table 3**
Effective odds ratios for and rank for the best PGF classification model. Only variables for which Z-score >2 are shown in this table.

| Variable | Odds ratio | Rank |
|---|---|---|
| | Geometric mean (2.5%, 97.5% percentiles) | Z-score |
| **Receiver data** | | |
| Female gender | 1.09 (0.95, 1.38) | 2.27 |
| Age | 1.03 (0.95, 1.11) | 2.50 |
| Height | 0.92 (0.82, 1.00) | 4.08 |
| Diagnosis | | 7.60 |
| ICM | 1.17 (0.96, 1.63) | |
| NICM | Base | |
| Congenital | 1.70 (1.24, 2.80) | |
| Graftfailure | 1.08 (0.85, 1.34) | |
| Misc | 1.00 (0.76, 1.40) | |
| Valve | 1.62 (1.19, 2.41) | |
| PVR | 1.12 (1.00, 1.26) | 4.94 |
| Antiarrhythmics | 1.14 (1.00, 1.46) | 2.76 |
| Most recent creatinine | 1.09 (1.02, 1.18) | 5.18 |
| Bilirubin | 1.06 (1.02, 1.10) | 3.39 |
| Medical condition | | 2.40 |
| Home | Base | |
| Hospital | 0.93 (0.69, 1.17) | |
| ICU | 1.05 (0.74, 1.43) | |
| Inotropic support pre-transplant | 0.84 (0.68, 0.98) | 2.72 |
| Ventilator | 1.90 (1.29, 3.07) | 7.44 |
| ECMO | 1.80 (1.25, 2.81) | 4.35 |
| Prior transplantation | 1.46 (1.16, 2.43) | 3.70 |
| Prior cardiac surgery | 1.24 (1.03, 1.66) | 4.58 |
| Pre-transplant blood transfusion | 1.23 (1.06, 1.47) | 4.13 |
| PRA >10% | 1.22 (1.02, 1.69) | 2.53 |
| **Donator data** | | |
| Age | 1.22 (1.05, 1.41) | 12.11 |
| Ischemic time (min) | 1.12 (1.04, 1.28) | 6.24 |
| CODD | | 3.95 |
| Head trauma | Base | |
| Other | 1.04 (0.76, 1.23) | |
| Stroke | 1.20 (1.00, 1.47) | |
| **Matching data** | | |
| HLA-B | | 2.86 |
| 0 mismatch | Base | |
| 1 mismatch | 1.08 (0.81, 2.07) | |
| 2 mismatch | 1.28 (0.97, 2.53) | |
| HLA-DR | | 2.22 |
| 0 mismatch | Base | |
| 1 mismatch | 1.01 (0.63, 1.37) | |
| 2 mismatch | 1.03 (0.63, 1.48) | |
| Gender | | 6.92 |
| Female to male | 1.34 (1.05, 1.80) | |
| Male to female | 0.94 (0.73, 1.21) | |
| Match | Base | |
| Weight (rec/don) | | 3.08 |
| 0.70–0.85 | 1.00 (0.69, 1.31) | |
| 0.85–1.13 | Base | |
| 1.13–1.30 | 1.06 (0.83, 1.47) | |
| <0.70 | 0.89 (0.64, 1.04) | |
| >1.13 | 1.26 (1.03, 1.70) | |

Variables included in: BUN, blood urea nitrogen; CODD, cause of donor death; CVI; ECMO, extracorporeal membrane oxygenation; HLA, human leucocyte antigen; ICM, ischemic cardiomyopathy; ICU, Intensive care unit; NICM, non-ischemic cardiomyopathy; PRA, panel reactive antibody level; PVR, pulmonary vascular resistance.

## 4. Discussion

We present a machine learning model that predicts the risk of PGF, with an AUROC of 0.69. The discrimination is superior to the previously developed RADIAL model. We also report that the choice of imputation technique for variables with missing data can significantly affect the result.

There are currently about 15 different algorithms that predict survival after heart transplantation [19]. Most are optimized for one-year survival or long-term survival such as the IHTSA model. There are only a few models for predicting PGF [6,20]. Those are mostly developed on small cohorts and rarely validated on external data.

The five most significant variables in our model of PGF are also in the IHTSA model of one year survival. However, ischemic time

**Table 4**
Average validation and test AUROC values for different network models. MPI refers to model trained with multiple probability imputation, and MI refers to models trained with mean and extra category imputation.

| Hidden layers | Network size | AUROC | | | |
|---|---|---|---|---|---|
| | | **Validation (MPI)** | **Validation (MI)** | **Test (95% CI) (MPI)** | **Test (95% CI) (MI)** |
| **0** | – | 0.663 | 0.698 | 0.666 (0.657, 0.678) | 0.699 (0.688, 0.709) |
| **1** | 50 | 0.676 | 0.697 | 0.685 (0.671, 0.692) | 0.697 (0.685, 0.706) |
| **2** | 50–50 | 0.675 | 0.697 | 0.674 (0.664, 0.685) | 0.698 (0.681, 0.701) |

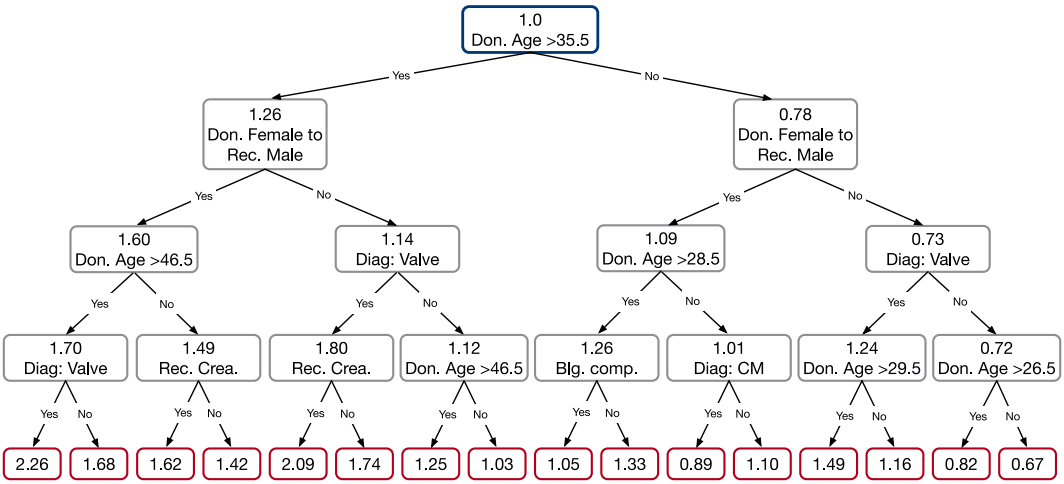


**Fig. 2.** Binary regression tree predicting the MLP model output (as odds) over the entire dataset. The most important variable is the top node, dark blue. Higher values indicate higher risks. The red boxes show the final odds. The Yes/No indicates the decision for the condition used by the regression tree. Sex is treated as a categorical value over the four possible combinations (female donor to male recipient, male to female, male to male and female to female), meaning that e.g. "female to male" is compared against the three other possible combinations. The depth of the tree was constrained during modeling to four levels to highlight the most important decisions. Blg, blood group; CM, cardiomyopathy; Crea, creatinine; Don, donor; Rec, recipient. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

and sex match were not among the 10 most significant variables in IHTSA [5]. The RADIAL model includes donor age and ischemic time. Both donor age and ischemia are known risk predictors of death after heart transplantation. Already in the early 1970s Griepp et al. suggested that an ideal heart donor should be younger than 30 years [21]. Our model also shows that prolonged ischemia time is associated with an increased risk of mortality, which was also seen in the RADIAL study. That ischemia does not end up as high in the IHSTA model's ranking is probably because that model is not primarily optimized for early graft failure but long-term survival time. An interesting finding from the present study is that sex mismatch is among the 5 top ranked risk factors. Conversely, interaction of the donor and the recipient weight was less important in this study. Pulmonary vascular resistance also ranked among the top 10 factors in this study [22]. Renal function came further down the ranking. Creatinine (or eGFR) is typically seen as a risk factor for 1-year survival but may be less important as a predictor of PGF.

At present, classical statistical methods are the most common way of identifying and ranking variables of clinical significance. Although many risk factors in connection with survival after transplants have been identified, the relationships between these variables (interactions) have not been sufficiently explored and especially not how this affects PGF [3,23]. Interactions between risk factors are complex and may be donor/recipient specific, as we have shown in previous studies of survival post heart transplantation [5,23].

Machine learning such as ANN has only been used in a few studies to predict long-term survival post heart transplantation [19]. However, in a relatively recently developed machine learning model, Tree of prediction (TOP), the authors show that 3-month mortality can be predicted with acceptable accuracy [24]. However, no external validation is available for that model.

The deep learning models can have an advantage here in the form that they include interactions and identify non-linear relationship in a more automated way. Although performance for our model does not reach the same levels as prediction models for 30 d mortality in cardiac surgery, the result was significantly better than for the RADIAL model [6,25]. One possible reason why deep neural network models did not improve classification performance further, not even in the presence of a large amount of data, is that variables in this and similar data cohorts have to some extent been pre-processed and managed. Most successes with deep networks have been with raw data, such as images or language data, from which representations and features must be identified by the learning algorithm.

An additional limiting factor is missing data [26]. We used a multiple imputation technique to use the complete dataset and avoid

selection bias if one would instead choose to remove patients or variables with missing values. However, this means that the significance for variables with many missing values is more difficult to quantify. We also tested using a missing values marker which is also an established method. For the present data cohort, this turned out not to be optimal because of a weak correlation between lack of data and poorer outcomes.

In order to assess how well a decision support can work in the clinical everyday life, it is important that it is correctly validated [26]. In medical specialties where the number of patients is not large enough, clinical databases are often required which collect information from many different centers. In this study, we have used the ISHLT Registry, the largest heart transplant database, which suggests generalizability. Furthermore, we have used a robust validation technique consisting of both cross-validation and cross-testing. Despite its limitations, multiple imputations greatly enhances generalizability because it reduces bias due to data missing not at random

The results of this study have limitations associated with the retrospective analysis of a registry database, quality of source data, and possible lack of standardization of some variables entered by multiple centers. Furthermore, the definition of primary graft dysfunction through registry data can be challenging. To minimize misclassification, we included severe forms of graft failure that resulted in death in the first 30 days after the transplant. We acknowledge that the availability of more granular data could provide a more nuanced approach to the diagnosis, but this was not possible using this registry data. The presence of missing data, especially if it is more common in one of the groups, is another problem that can affect the result even though we tried to minimize this impact using multiple imputation technology. While the AUROC of our model is superior to many previous studies, it is clear that additional factors that were not or cannot be accounted for at the time of transplant impact the risk of PGF.

In summary, we present an ANN based prediction model for PGF. Its discrimination capacity is better than that of other models in heart transplant that used more traditional statistical methods. Whether the improved discrimination model can assist in clinical decisions at the time of transplant should be further tested.

## Author contribution statement

Johan Nilsson and Mattias Ohlsson: Conceived and designed the experiments; Analyzed and interpreted the data; Wrote the paper.
Björn Linse: Performed the experiments; Wrote the paper.
Lars Lund and Bodil Andersson: Analyzed and interpreted the data; Wrote the paper
Josef Stehlik: Contributed reagents, materials, analysis tools or data; Analyzed and interpreted the data; Wrote the paper.

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## Data availability statement

The authors do not have permission to share data.

## Declaration of interest's statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Additional information

No additional information is available for this paper.

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